Daily Research Digest — 2026-04-26

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

What is new and worth noticing

Recent papers present intriguing findings on sterile neutrino decay signatures in galaxy clusters, conditions for accelerating scaling attractors without dark energy, and a novel framework to constrain primordial non-Gaussianity bias. There are also detailed analyses of long-term ionized outflows in the Seyfert 1 galaxy NGC 3783, providing insights into galactic dynamics.

Most interesting papers

- **An RXTE Search for the Sterile Neutrino Decay in Galaxy Clusters**
- Found a 2.5σ excess of emission over thermal models at 3.6 keV in galaxy clusters, indicative of sterile neutrino decay with a significant correlation to cluster temperature. This suggests that dark matter decay rates could be linked to the thermodynamic properties of clusters.
- **Accelerating scaling solutions from dark matter particle creation**
- Identified conditions under which adiabatic dark matter particle creation can mimic dark-energy-like behavior, leading to accelerating scaling attractors. The interaction must depend on dark matter density and energy flow from DM to a barotropic fluid.
- **Informative Priors on Primordial Non-Gaussianity Bias b_ϕ From Galaxy Formation**
- Developed a framework to construct physically motivated priors for the bias parameter b_ϕ, critical for constraining primordial non-Gaussianity. This addresses systematic errors in current analyses and provides more robust inference of $f_{\mathrm{NL}}^{\mathrm{loc}}$.
- **Delving into the depths of NGC 3783 with XRISM: V. Broad-band modeling of ionized outflows**
- Analyzed long-term X-ray spectra from NGC 3783, identifying eight warm absorber components with consistent column densities and turbulent velocities over two decades but noting a decrease in low-ionization components.

Findings by theme

Dark Matter and Galaxy Clusters

- The RXTE search for sterile neutrino decay in galaxy clusters has found significant evidence of an excess emission at 3.6 keV, correlating strongly with cluster temperature and suggesting a link between dark matter decay rates and thermodynamic properties.

Galaxy Formation and Primordial Non-Gaussianity

- A new framework constructs physically motivated priors on the bias parameter b_ϕ, addressing systematic errors in constraining primordial non-Gaussianity via galaxy clustering data. This approach uses CAMELS-SAM simulations to derive unbiased constraints.

Cosmological Dynamics and Accelerating Scaling Attractors

- Conditions for accelerating scaling attractors without dark energy have been identified through models where dark matter particle creation is controlled by DM density, leading to coexistence of fluids with fixed energy fractions at late times. This demonstrates a mechanism for late-time acceleration without invoking dark energy.

Galactic Outflows and X-ray Spectroscopy

- Long-term X-ray observations of NGC 3783 reveal stable warm absorber components with consistent physical properties over two decades, but show a reduction in low-ionization components compared to earlier data.

Connections and tensions

These findings tie into ongoing debates about dark matter decay signatures, the conditions for accelerating scaling attractors without invoking dark energy, and the impact of galaxy formation processes on non-Gaussianity constraints. The results from NGC 3783 provide observational support for theoretical models of ionized outflows and their long-term stability.

Open questions and follow-ups

- Confirming the sterile neutrino decay signal in galaxy clusters with higher statistical significance to understand its implications for dark matter properties.
- Investigating the conditions under which adiabatic particle creation can fully mimic dark energy behavior, potentially reducing reliance on exotic dark energy models.
- Extending the framework for constructing priors on b_ϕ to other types of galaxies and observational datasets to validate its robustness across different contexts.
- Further monitoring and modeling ionized outflows in AGN environments over extended periods to understand their evolutionary dynamics and connection to galaxy feedback mechanisms.

Paper Summaries and Figures

1. An RXTE Search for the Sterile Neutrino Decay in Galaxy Clusters

Focus: galaxy clusters, galaxies, dark matter
Authors: Mark J. Henriksen
Published: 2026-04-23T16:11:53+00:00
PDF: https://arxiv.org/pdf/2604.21817v1
Summary: We have searched for the 3.55 keV line from sterile neutrino decay using 3.1 megaseconds of RXTE cluster data. A 2.5σ excess of emission over a thermal model is found over the energy span of the 3.55 keV line in the combined spectra of the eight clusters that individually have an excess. The residuals are added to increase the signal to noise ratio of the excess, which is then modeled with a Gaussian to simulate the instrumental spectral response. We find a significant correlation (r = 0.76) for a line centered at 3.6 keV with a model flux of 3.07 x 10^{-5} ph cm^{-2} s^{-1}. Mixing angle for detected clusters ranges from 0.35 to 6.2 x 10^{-10}. The decay rate inferred from the line flux is strongly correlated (r = 0.87) with cluster temperature, which is due to hotter, more massive clusters having a larger amount of dark matter. Approximately half of the decay line total flux comes from the Coma cluster. We fit the Coma cluster spectrum with two different three-component models. The first includes a Gaussian fixed at 3.55 keV to model soft emission. The second three-component model uses a second thermal component to model soft emission. The model fit parameters indicate that the second thermal component is modeling high-energy residuals rather than low ones, where the Gaussian is. Though our line fluxes exceed most reported detections and upper limits, they do not overproduce the dark matter. We conclude that some fraction of the marginally detected excess could be attributed to the decay line since low-temperature thermal emission and systematics fail to model it completely.

Selected figures

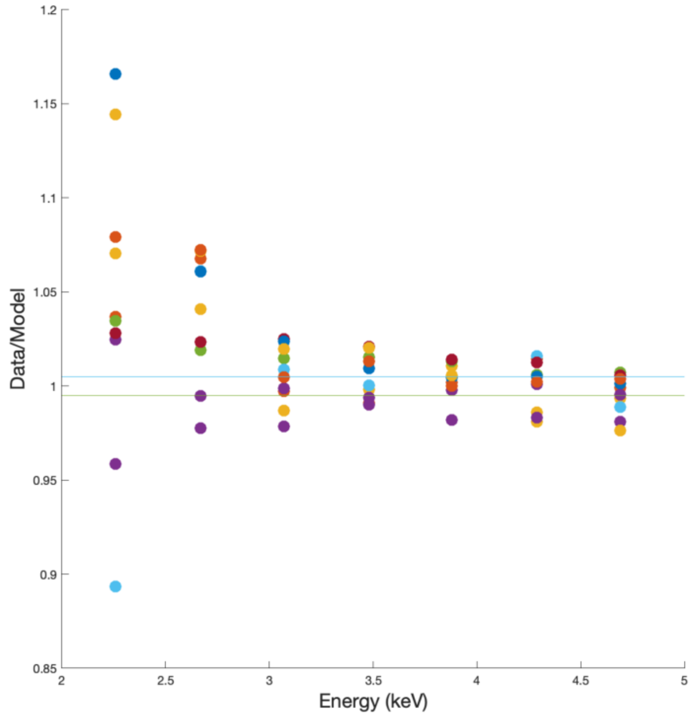

Figure 3. The ratio (data/model) in each channel covering the energy range of the 3.55 k feature folded through the RXTE PCA energy resolution is shown for each cluster. Lines a that bracket the systematic error for the PCA as ś0.5%. The features can be compared systematic error and the number of c

2. Informative Priors on Primordial Non-Gaussianity Bias b_ϕ From Galaxy Formation

Focus: galaxies
Authors: Anne Moore, Lucia A. Perez, Elisabeth Krause
Published: 2026-04-23T15:46:53+00:00
PDF: https://arxiv.org/pdf/2604.21790v1
Summary: Constraining primordial non-Gaussianity via its scale-dependent imprint on galaxy clustering requires knowledge of the bias parameter b_ϕ, which is exactly degenerate with $f_{\rm NL}^{\rm loc}$ at leading order. To break this degeneracy, current analyses adopt the relation $(b_\phi = 2\delta_c\,(b_1 - 1))$ based on the assumption of a universal mass function. This relation is known to break down for physically motivated galaxy selections, introducing systematic errors in the inferred $f_{\rm NL}^{\rm loc}$ that scale directly with the assumed b_ϕ prior. We present a framework to construct physically motivated, observation-conditioned priors on b_ϕ by marginalizing over galaxy formation uncertainties. We use the CAMELS-SAM simulation suite, augmented by separate Universe simulations, to measure galaxy formation observables, like the stellar mass function (SMF) and the stellar-to-halo mass relationship (SHMR), and b_ϕ across a range of galaxy formation parameters. From these measurements, we construct a distribution of b_ϕ conditioned on observations, and we select our galaxy sample to resemble the DESI Emission Line Galaxy (ELG) sample. Conditioning on the SMF or SHMR decreases σ_{b_ϕ} from 0.69 to 0.08 and 0.02 respectively – reductions of 88% and 97% – with consistent results when conditioning on the observed data directly. Despite substantial shifts in the galaxy formation posteriors driven by known SC-SAM discrepancies at high halo masses, the resulting b_ϕ distributions remain mutually consistent across all observables. The SMF and SHMR are found to carry sufficient constraining power to reduce the galaxy formation uncertainty in b_ϕ relevant for $f_{\rm NL}^{\rm loc}$ inference with next-generation spectroscopic surveys

3. Accelerating scaling solutions from dark matter particle creation

Focus: dark matter
Authors: Sudip Halder, Jaume de Haro, Supriya Pan, Emmanuel N. Saridakis, Tapan Saha, Subenoy Chakraborty
Published: 2026-04-23T15:28:34+00:00
PDF: https://arxiv.org/pdf/2604.21770v1
Summary: This article opens new window to obtain accelerating scaling attractors without any need of dark energy. We study cosmological dynamics in a two-fluid system where pressureless dark matter (DM) undergoes adiabatic particle creation and exchanges energy with a barotropic fluid. Considering six widely used interaction prescriptions, we formulate the corresponding autonomous systems in a compact phase space and perform a unified dynamical analysis. We find that accelerating scaling attractors, namely late-time states where both fluids coexist with fixed energy fractions, arise only when the interaction is controlled by the DM density and energy flows from DM to the second fluid. Such attractors appear in the global and local DM-based interactions, and in the global mixed case, but are entirely absent when the interaction depends on the second fluid or on local mixed terms, which instead drive the universe to a DM-dominated accelerating phase. These results clarify the unique conditions under which matter creation can mimic dark-energy-like behaviour without introducing a dark-energy component.

Selected figures

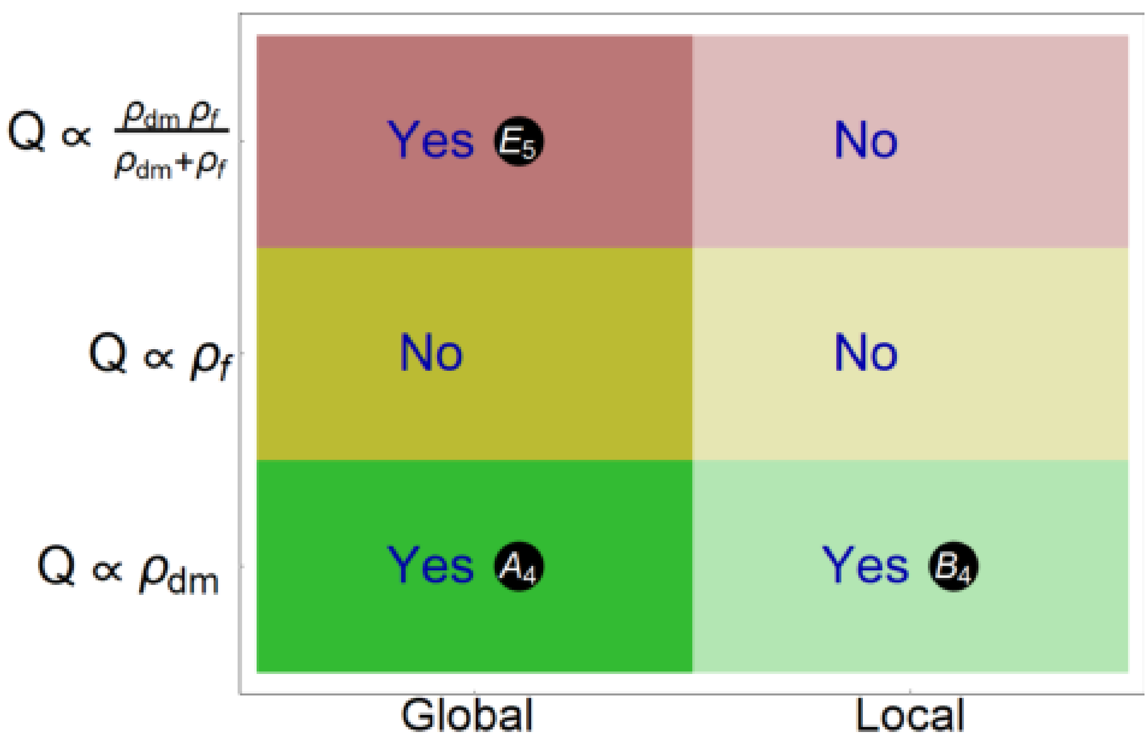

FIG. 7. Summary of accelerating scaling attractors identif in this study. Each row corresponds to a particular ene density (green, yellow, or pink), while each column represe the type of interaction: global rate (dark shading) or lo rate (light shading).

4. Delving into the depths of NGC 3783 with XRISM: V. Broad-band modeling of ionized outflows

Focus: galaxies
Authors: Keqin Zhao, Jelle S. Kaastra, Liyi Gu, Missagh Mehdipour, Megan E. Eckart, Keigo Fukumura, Matteo Guainazzi, Chen Li, Christos Panagiotou, Matilde Signorini
Published: 2026-04-23T14:16:59+00:00
PDF: https://arxiv.org/pdf/2604.21710v1
Summary: The Seyfert 1 galaxy NGC 3783 hosts a multiphase warm absorber (WA) that has been extensively studied in the X-ray band. High-resolution spectra from 2000-2001 revealed a complex outflow with multiple ionization and velocity components. Two decades later, new XMM-Newton and XRISM observations allow us to investigate the long-term evolution of these outflows. We perform joint spectral modeling of the XMM-Newton/RGS and XRISM/Resolve time-averaged spectra using the pion photoionization code within SPEX. We derive the ionization parameter, column density, turbulent velocity, and outflow velocity for each absorption component, and investigate their thermal stability and Absorption Measure Distribution (AMD) to characterize the physical and dynamical properties of the WA in NGC 3783 in 2024. We compare these results with the 2000-2001 epoch to assess long-term variability, stability, and possible changes in the absorber population. We identify eight WA components spanning log = 1.08-3.38 and outflow velocities of 480-1230 km s^{-1}. The ranges of column densities and turbulent velocities remain broadly consistent with the WAs from 2000-2001, but the earlier data contained more low-ionization, high-velocity components. The total column density in 2024 is 1.5 times larger than in 2000-2001, requiring replenishment by fresh material. The dominant Unresolved Transition Array (UTA) absorber (Comp. B3) has increased its column density by a factor of three while maintaining a similar ionization parameter. The WAs in NGC 3783 have undergone significant structural and dynamical evolution over the past 24 years.

Selected figures

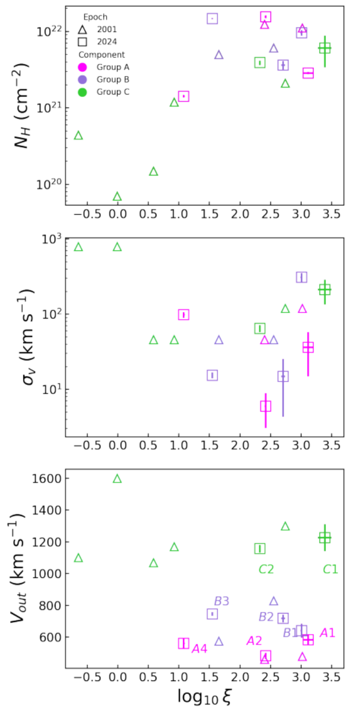

Fig. 3. Relations between the parameters of the eight outow compo- nents derived from the XRISM Resolve spectrum of NGC 3783. The top panel displays the column density NH as functions of the ionization parameter (log). The middle and bottom panels show the turbulent ve- locity (\sigmav) and outow vel

5. Saturation Mechanisms in the Interacting Dark Sector

Focus: dark matter
Authors: Andronikos Paliathanasis, Kevin J. Duffy
Published: 2026-04-23T13:36:31+00:00
PDF: https://arxiv.org/pdf/2604.21671v1
Summary: We introduce a family of phenomenological cosmological models featuring an interacting dark sector modulated by a sparseness scale parameter, in order to describe the late-time accelerated expansion of the universe. The sparseness scale, inspired by well-established saturation mechanisms in ecology and biology, is introduced in the interaction as a half-saturation constant that bounds the energy exchange between dark matter and dark energy, controls the dynamical behaviour of the physical variables and can prevent the phantom crossing. We consider three nonlinear interacting models, where two of them recover the linear interacting scenarios when the sparsity parameter vanishes. We examine the phase-space of the cosmological field equations by using the Hubble normalization approach. We determine the stationary points and their stability properties in order to reconstruct the asymptotics behaviour of the field equations. Such an analysis allows us to demonstrate the effects of the sparseness scale on the background dynamics. We test the interacting models with observational data. Specifically, we employ Supernovae catalogues, cosmic chronometers, Baryon Acoustic Oscillation measurements from DESI DR2, and redshift-space distortion measurements of the growth of large-scale structure through the f and $f\sigma_8$ observables. The Bayesian analysis suggests that, for two of the three models, a vanishing sparsity parameter is disfavoured at more than the 95\% confidence interval, providing observational support for a nonzero sparseness scale in the dark sector interaction.